

Payments, Gateways and Security

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Architecture

Buyer → PSP-hosted/tokenized checkout → provider server confirmation/webhook → Vossie verifies signature/idempotency → ledger entry/state transition → protection window → payout.

Vossie should store

- internal payment ID,
- provider reference,
- amount/currency,
- fee snapshot,
- status,
- seller/courier allocation,
- refund/payout references.

Vossie should not store

- raw card number,
- CVV,
- online-banking credentials.

Prototype methods to demonstrate

- card,
- Instant EFT / Pay by Bank,
- PayShap,
- Apple Pay / Google Pay where provider/device appropriate,
- sandbox only unless real credentials are provided.

Security controls

- backend-only secret keys,
- signed webhook verification,
- idempotent event processing,
- payout-detail changes protected with step-up security,
- reconciliation jobs,
- append-only financial history,
- client never declares final payment success by itself.

Provider strategy

Use a PSP adapter. Candidate providers were researched across Paystack, Payfast, Peach, Stitch/Ozow and others. Final production choice remains open.